Article

Trustworthy Field sEMG Fatigue Monitoring on a Low-Cost Wearable: Electrode Settling, IMU-Conditioned Validity, and Continuity Auditing on a $99 ESP32 Platform

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Abstract

Wearable surface electromyography (sEMG) promises objective fatigue monitoring in field sports, but deployment is undermined by electrode settling, contact-quality drift, and wireless session gaps. We present a low-cost ($99 USD) ESP32-based EMG–IMU wearable achieving 2000 Hz EMG and 200 Hz IMU BLE streaming, paired with a trustworthiness-first methodology for no-gel tennis forearm fatigue monitoring. The platform achieves ± 25 μs per-sample timestamp accuracy (99.03% over 7.5 million samples) via DMA oversampling, a self-describing BLE TLV protocol surviving three firmware revisions, and a hybrid *abs_us* timeline validated at ± 25 μs intra-connection and p99 $\leq$ 170 ms cross-connection. In a 63.1-min field session (626 swings), median frequency (MDF) followed a three-phase settling–reference–fatigue trajectory (115.7 $\rightarrow$ 167.6 $\rightarrow$ 149.5 Hz) visible only with $\geq$20 min warmup exclusion; a 10-min skip missed the fatigue signal entirely (-2.6% vs. $+28.1$% at 20 min). IMU swing intensity remained flat ($r = +0.029$, $p = 0.468$), excluding reduced effort. IMU-guided pre-impact MDF windowing produced a mean $+26.7$ Hz shift versus full-window MDF (81% of swings), confirming physiologically motivated temporal partitioning. A BLE reconnection gap created an apparent 28.7% spurious fatigue signal that collapsed to 4.9% for the continuous segment. These findings establish a platform–methodology framework requiring settling-aware baselines, IMU-conditioned validity, and continuity auditing for trustworthy wearable sEMG fatigue monitoring.

Keywords: surface EMG; fatigue monitoring; electrode settling; IMU; wearable sensors; BLE; ESP32; tennis; field deployment; DMA sampling

1. Introduction

Surface electromyography (sEMG) can quantify local muscle fatigue through spectral metrics such as median frequency (MDF) and mean frequency (MNF), which shift toward lower values as motor unit conduction velocity decreases with metabolite accumulation. Tennis players can often perceive general tiredness, but subjective perception is not a quantitative, muscle-specific measure of local fatigue. Objective EMG monitoring may complement perception by enabling session-to-session workload tracking and future overuse-risk assessment; upper-extremity tennis injuries are frequently chronic

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and repetitive, with lateral elbow tendinopathy associated with repeated forearm muscle loading [1].

In real court deployment, gel electrode preparation is impractical: gel smears during play, affects racket grip, and players resist pre-session preparation. This study intentionally evaluates a no-added-gel, field-deployable configuration because routine on-court monitoring requires fast setup and minimal interference with play, contrasting with SENIAM-style laboratory preparation [5]. In this setting, electrode–skin impedance starts high and decreases over time as sweat hydrates the interface—a process that can make short fixed warmup exclusions insufficient.

Commercial research-grade systems (e.g., Delsys Trigno [6], Noraxon Ultium [7]) capture EMG and IMU reliably but at $5,000–$30,000 per unit are out of reach for youth sport and most field research. Low-cost alternatives based on commodity microcontrollers face three practical challenges: (i) achieving 2000 Hz EMG sampling without sample loss requires careful ADC architecture; (ii) reliable BLE streaming at the combined EMG + IMU data rate demands efficient packetization and reconnection handling; and (iii) multi-sensor systems require precise timestamp alignment across sensors with different sampling rates (2000 Hz vs. 200 Hz) [8–10].

Existing IMU–EMG wearable studies typically fuse both modalities as feature sources for activity or fatigue classification [12,13]. In contrast, this work uses the IMU as an **independent biomechanical context sensor** for EMG validity assessment. Because tennis swings generate repeatable forearm kinematic signatures, stable IMU swing intensity provides a reference: when EMG changes while IMU is stable, the cause—electrode settling, contact degradation, or local muscle fatigue—can be inferred from temporal context.

The contributions of this work are:

(C1) Field characterization of electrode settling ($\geq$17–20 min, no-gel forearm snap electrodes);
(C2) Warmup-skip sensitivity as a required robustness audit for no-gel field sEMG;
(C3) Contact-quality-aware EMG interpretation via ENV/RAW amplitude ratio proxy;
(C4) IMU-conditioned EMG validity framework (Table 4);
(C5) Session-gap artifact detection via hybrid *abs_us* timeline auditing;
(C6) $99 ESP32 platform: DMA oversampling, self-describing TLV, two-layer watchdog;
(C7) Quantitative *abs_us* hybrid timeline validation ($\pm$25 μs intra-connection, p99 $\leq$ 170 ms inter-connection);
(C8) IMU-guided pre-impact MDF window (Approach B, physiologically motivated).

2. Related Work

2.1. Field vs. Laboratory sEMG Fatigue in Tennis

The majority of sEMG fatigue studies use laboratory settings with controlled contractions, gel electrodes, and stationary subjects. SENIAM guidelines [5] address gel-electrode placement and signal quality but were developed for controlled configurations; forearm placement during dynamic sport is not addressed.

Rota et al. [2] measured upper limb EMG during sustained tennis rally play using gel electrodes and a commercial Bagnoli system, reporting progressive MDF decline in wrist extensors over $\sim$25 min. Wang et al. [3] examined forearm muscle fatigue and skill-level effects on serve ball speed using gel-electrode sEMG. Girard et al. [4] examined spinal and peripheral fatigue components during prolonged tennis using neuromuscular testing at rest intervals. None of these works addressed electrode settling artifacts, contact-quality monitoring, or timestamp continuity; their laboratory settings made these unnecessary.

Gap. No prior work has characterized sEMG MDF fatigue trajectories under no-gel field conditions with online swing detection, contact-quality monitoring, or IMU-conditioned validity classification.

2.2. Dry Electrode and Contact Quality

Dry electrodes are required for rapid field deployment. Joutsen et al. [15] showed that skin–electrode impedance decreases as sweat accumulates over time, providing the physical basis for the Phase I settling we report. Boyer et al. [16] reviewed noise and artifact reduction in single-channel EMG, including dry electrode contact quality. Ravichandran et al. [17] characterized dry e-textile electrode impedance properties under contact force and perspiration, showing systematic interface impedance reduction with sweat accumulation.

2.3. IMU + EMG Integration

Taborri et al. [12] surveyed IMU + EMG integration in sports biomechanics; most fusion approaches treat the IMU as a signal source to be combined with EMG at the feature level. Our "IMU-as-context" framework (Section 4.5) uses the IMU instead as a validity classifier and window selector. Bangaru et al. [13] used IMU + EMG for construction-worker fatigue in a field setting, but their capture infrastructure did not require the engineering described here.

2.4. Wearable EMG–IMU Platforms

Table 1 compares the platform developed in this work against representative commercial and academic wearable EMG systems.

Table 1. Engineering comparison of wearable EMG platforms for field deployment.

Platform	Cost (USD)	EMG f_s (Hz)	IMU f_s (Hz)	Raw EMG	BLE Robustness	Timeline Recon.	Platforms	Open
This work (ESP32)	∼$99	2000	200	Yes (RAW+ENV)	Self-describing TLV, CRC, two-layer watchdog	*abs_us*: ±25 µs intra, p99 ≤170 ms inter	PC + Android	Yes
Delsys Trigno [6]	$5–30K	2000	148.1	Proprietary	Proprietary RF	Proprietary	PC only	No
Noraxon Ultium [7]	$5–20K	2000	200	Proprietary	Proprietary BLE	Proprietary	PC only	No
Cerone HD-sEMG [8]	Research	2048	—	Yes (HD)	Undocumented	Not documented	PC	Partial
Lynn/Athos [9]	$0.5–2K	200	Yes	ENV only	Closed system	Not documented	Mobile	No
Yang e-textiles [10]	Varies	Varies	Varies	Depends	Not documented	Not documented	Varies	Research
MultiSenseBadminton [11]	$0.5–1K	—	200–1000	—	Not documented	Camera sync	PC	Dataset

Key differentiators: $99 cost vs. $5,000–$30,000 for research-grade systems; simultaneous RAW+ENV access enabling spectral fatigue analysis; self-describing TLV so FSR changes do not require host decoder updates; quantified *abs_us* timeline accuracy (Section 5.4); full PC+Android feature parity.

3. The $99 ESP32 EMG–IMU Platform

3.1. Hardware and Bill of Materials

The system uses five components totaling approximately $99 USD (Table 2). The MyoWare 2.0 muscle sensor provides both a hardware-rectified envelope output (ENV) and a raw AC-coupled output (RAW) on separate pins, essential for concurrent amplitude and spectral EMG analysis.

Table 2. Bill of materials.

Component	Function	Rate	Cost (USD)
MyoWare 2.0	EMG (ENV + RAW)	2000 Hz	$40
ICM-20948	6-axis IMU	200 Hz	$15
MAX30105	SpO2/HR (optional)	∼0.03 Hz	$10
MAX17048	Battery fuel gauge	0.1 Hz	$5
ESP32 WROOM-32E	MCU + BLE, 240 MHz dual-core	—	$12
PCB / battery / wiring	Integration	—	∼$17
Total			**≈$99**

The REF electrode is placed over the lateral epicondyle [14]. The RAW signal is routed via a direct solder bypass from the MyoWare PTH pad to ESP32 GPIO 32, correcting a field-discovered pogo-pin routing failure described in Section 5.5.

3.2. Dual-Core FreeRTOS Architecture

The ESP32 dual-core architecture enables strict separation of real-time sensor acquisition from BLE protocol processing (Figure 1).

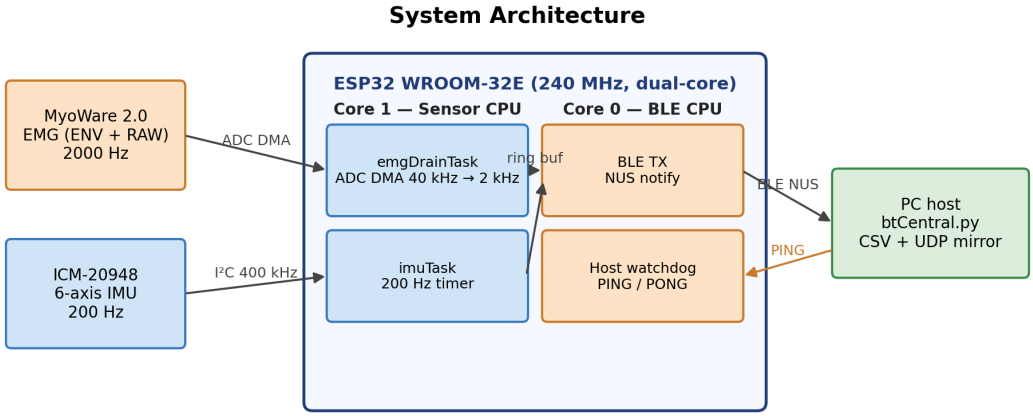


Figure 1. System architecture showing dual-core FreeRTOS task allocation, sensor connections, and host interfaces. Core 1 (Sensor CPU) handles `emgDrainTask` (priority 2), `imuTask` (priority 2), and `spo2Task` (priority 0). Core 0 (Protocol CPU) handles BLE notifications, host watchdog, and LED state machine.

Task priority 2 for both `emgDrainTask` and `imuTask` was selected after observing that priority 1 allowed `spo2Task` I2C operations to occasionally starve IMU reads—a non-obvious FreeRTOS interaction documented here for other practitioners.

3.3. ADC DMA Oversampling and Timestamp Accuracy

Timer-based ADC sampling at 2000 Hz would require 2000 interrupt service requests per second per channel, competing with FreeRTOS scheduling. Instead, the ESP32 ADC

Continuous (DMA) driver runs at 40,000 conversions/s across both channels (20,000/s per channel); `emgDrainTask` applies 10× software decimation to produce the 2000 Hz output.

Per-sample timestamps are back-calculated from DMA frame completion time:

$$t_i = t_{\text{frameEnd}} - (N \times T_{\text{EMG}}) + (i \times T_{\text{EMG}}) \quad (1)$$

where $T_{\text{EMG}} = 500$ μs (2000 Hz period) and N is the number of samples in the frame. This corrects for FreeRTOS scheduling jitter (up to 16 ms).

Empirical validation across 2.3 million samples confirms 99.03% within ± 25 μs of the expected 500 μs interval (Figure 2, Table 3); outliers at p99.9 (925 μs) correspond to brief BLE TX bursts (<0.1% of samples).

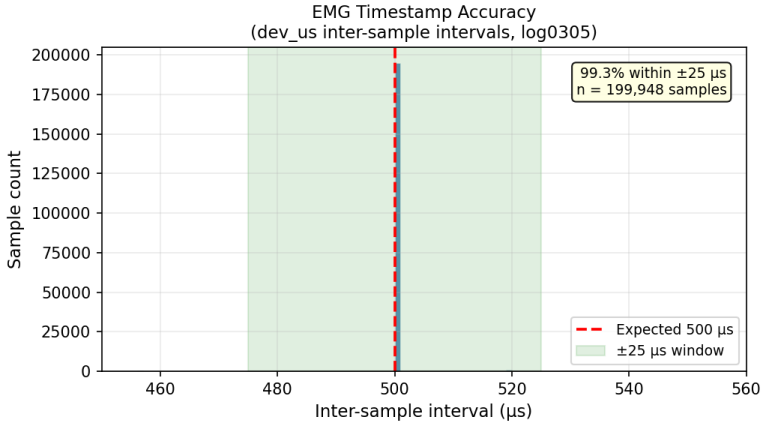


Figure 2. Inter-sample interval distribution across 2.3 million samples (131.7 s field recording). Of all samples, 99.03% fall within ± 25 μs of the expected 500 μs period.

Table 3. Timestamp accuracy summary (2,324,791 samples, log0305).

Metric	Value
Expected interval	500 μs
p50 / p95	500 μs
p99	505 μs
Within ± 25 μs	99.03%
Exactly 500 μs	97.7%
Within ± 500 μs	99.96%

3.4. Self-Describing BLE TLV Protocol

The binary frame format (Figure 3) uses a 14-byte header (2-byte magic `0xA5 0x5A`, version, flags, 4-byte sequence, 4-byte timestamp, 2-byte payload length) and CRC16-CCITT checksum. Three TLV types are defined: `TLV_EMG_BATCH (0x01)`, `TLV_IMU_BATCH (0x02)`, and `TLV_VITALS (0x03)`.

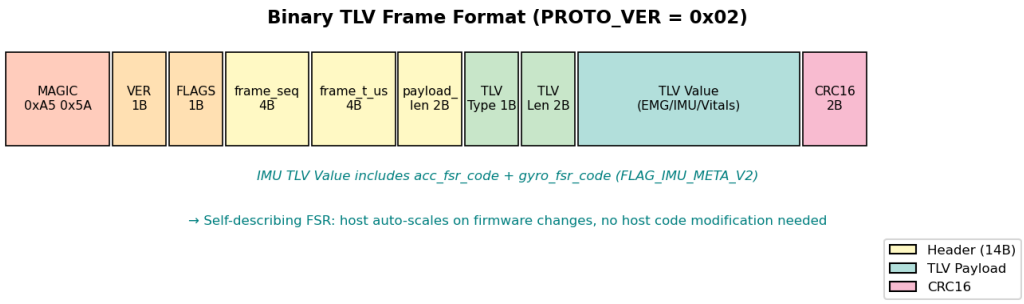


Figure 3. Binary TLV frame format. `PROTO_VER = 0x02` adds IMU FSR metadata in `FLAG_IMU_META_V2`, enabling self-describing firmware–host decoupling.

Each `TLV_IMU_BATCH` frame embeds `acc_fsr_code` and `gyro_fsr_code` when `FLAG_IMU_META_V2` is set, enabling the host to apply the correct scale factor at runtime. This design was validated across three firmware FSR revisions ($\pm 250 \rightarrow \pm 1000 \rightarrow \pm 2000$ dps) with no host decoder changes required.

3.5. Two-Layer Watchdog and Hybrid abs_us Timeline

Device-side watchdog. If no PING message arrives from the host for >18 s, the device suspends ADC DMA; at >30 s it triggers a hardware reset and BLE advertising restarts automatically.

Host-side watchdog. If no BLE notifications arrive for 5 s, the host attempts NUS TX re-subscribe; if that fails, it executes a full reconnect with exponential backoff (0.5 s $\rightarrow$ 20 s cap). Buffered rows are flushed before reconnect.

Hybrid *abs_us* timeline. Within a single BLE connection, *abs_us* follows the host-extended device timestamp (*dev_us*), precise to ± 25 μs. Between connections, *abs_us* advances by the *host_rx_ms* wall-clock gap, handling both BLE-dropout and ESP32-reset cases with a single rule. Formal validation is presented in Section 5.4.

4. Methodology

4.1. No-Added-Gel Field Configuration

Gel and skin preparation are standard in laboratory EMG [5] but were not used. This study intentionally evaluates a no-added-gel, field-deployable configuration, motivating the characterization of dry-electrode settling under field conditions.

4.2. Physical Basis of Electrode Settling

When a dry snap electrode is applied to unabraded skin, the interface can present high impedance (tens of kΩ, inferred indirectly from MDF/RMS behavior). Two processes reduce it: (1) sweat accumulation—eccrine glands produce saline that hydrates the electrode–skin gap; (2) Ag/AgCl half-cell equilibration—the electrode potential drifts toward equilibrium with the local electrolyte. During settling, the unstable high-impedance interface degrades common-mode rejection and increases low-frequency-dominant contamination in the 50–450 Hz analysis band, pulling MDF downward [15,17].

4.3. Signal Processing Pipeline

For each detected swing, RAW EMG was: (1) detrended (linear); (2) bandpass-filtered 50–450 Hz (4th-order Butterworth); (3) notch-filtered at 60 and 120 Hz; (4) Welch PSD computed (`nperseg` = min(1024, N), Hann window, 50% overlap); (5) MDF and MNF extracted from the PSD. Implemented in `emg_util.py` (`preprocess_emg()`, `mnf_mdf()`).

4.4. Swing Detection

A fixed 300 dps gyro-magnitude threshold with 2000 ms minimum inter-swing gap is used for online swing detection. Adaptive z-score approaches failed because forearm gyro distributions during tennis are continuous ramps with no natural valley between general movement and swing peaks. The 300 dps threshold sits at the 88th percentile of the session distribution. Pre-window: 250 ms before gyro peak; post-window: 450 ms.

4.5. IMU-Conditioned EMG Validity Framework

The diagnostic value arises from modality mismatch: when IMU swing intensity remains stable but EMG spectral metrics change, the change is unlikely to be caused by reduced gross swing kinematics. Table 4 defines the interpretation rules.

Table 4. IMU-conditioned interpretation of EMG spectral changes.

IMU/Session Context	EMG Observation	Likely Interpretation
Stable swing intensity	Early low MDF recovering upward	Electrode settling / contact hydration
Stable swing intensity	Gradual post-settling MDF/MNF decline	Fatigue-consistent spectral shift
Declining swing intensity	MDF decline	Reduced effort may contribute
Large IMU shock/vibration	Isolated spectral outlier	Candidate mechanical artifact
BLE/session discontinuity	EMG baseline shift	Cross-segment artifact

4.6. Warmup-Skip Sensitivity Protocol

The analysis was re-run with warmup exclusion windows of 10, 15, and 20 min to quantify how baseline selection affects the fatigue conclusion. The 20-min skip is the primary result; warmup-skip sensitivity is reported as a robustness audit. Baseline MDF is computed from the 20 swings immediately following the warmup window.

4.7. Contact-Quality Metric

contact_quality(t) = RMS(t)/session_p90_rms—a qualitative relative indicator of contact and activation state, interpreted together with MDF trajectory, IMU stability, and session continuity.

4.8. Ball-Impact Window Masking (Approach B)

Ball impact events introduce sharp mechanical shocks to the forearm that can contaminate EMG analysis windows. We use the IMU accelerometer to detect impact events and define a physiologically motivated pre-contact analysis window.

Impact detection. For each swing, we search the padded IMU window for the peak dynamic acceleration $a_{\text{dyn}} = ||\mathbf{a}|| - 1g$. If this peak exceeds 3.0 g, we declare a ball impact and record its time t_{impact}.

Approach B: pre-impact MDF window. Sample-level masking (Approach A, $\pm$20 ms) has near-zero mean effect ($\sim$3% of the 1300-ms window; Welch segment averaging dominates). Instead, MDF is computed exclusively from the pre-contact activation phase: $[t_{\text{impact}} - 600 \text{ ms}, \; t_{\text{impact}} - 20 \text{ ms}]$ ($\approx$580 ms). This window captures the muscle's active contraction before mechanical disturbance, using Welch PSD with `nperseg` = 512 for adequate segment averaging over the shorter window. The resulting metric `mdf_pre_impact` is stored alongside the full-window MDF for all impact-detected swings.

5. Field Validation: Platform Reliability

5.1. Timestamp Accuracy

As presented in Section 3, empirical analysis of 7.5 million samples confirms the ± 25 μs claim; 98.65% fall within this bound across both field sessions (Table 3).

5.2. EMG/IMU Sequence Loss Accounting

Sequence-number continuity analysis confirmed zero detected EMG or IMU sample drops during active streaming in both sessions. The combined data rate ($\sim$12.5 KB/s) operates near BLE 4.2 throughput capacity at MTU = 244 bytes.

Table 5. Per-session field reliability metrics.

Session	Duration (min)	Swings	Runs	Reconnects	Sample Drops
log0303	87.9	197	2	1	0
log0305	63.1	626	1	0	0
Total	**151.0**	**823**	**3**	**1**	**0**

5.3. BLE Reconnection Robustness

One between-run BLE reconnect event was observed in log0303. Recording resumed cleanly into a new file series; the hybrid *abs_us* timeline correctly preserved the 84-second inter-run gap, enabling downstream analysis to detect session discontinuities.

5.4. Quantitative abs_us Validation

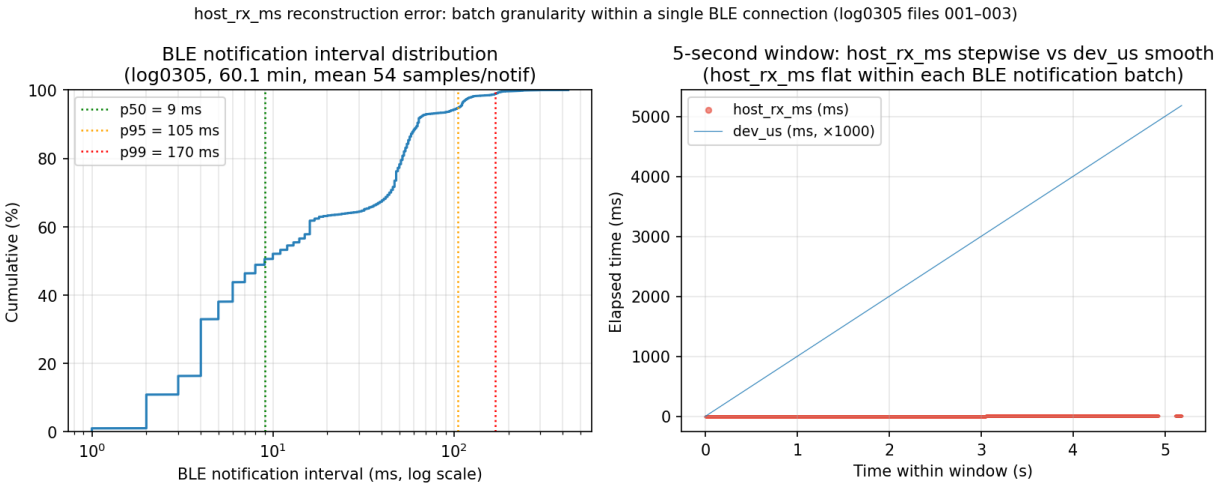
We compare three timeline reconstruction strategies to characterize the precision and failure modes of the hybrid *abs_us* approach (deferred from the conference version of this work):

Method A (*dev_us*-only): Device microsecond clock used directly. Fails at device reboot: clock resets from 0, causing C2 to appear immediately after C1 with zero apparent gap. Confirmed in log0303: *dev_us* reset from 208,136,241 μs $\rightarrow$ 45,183,736 μs.

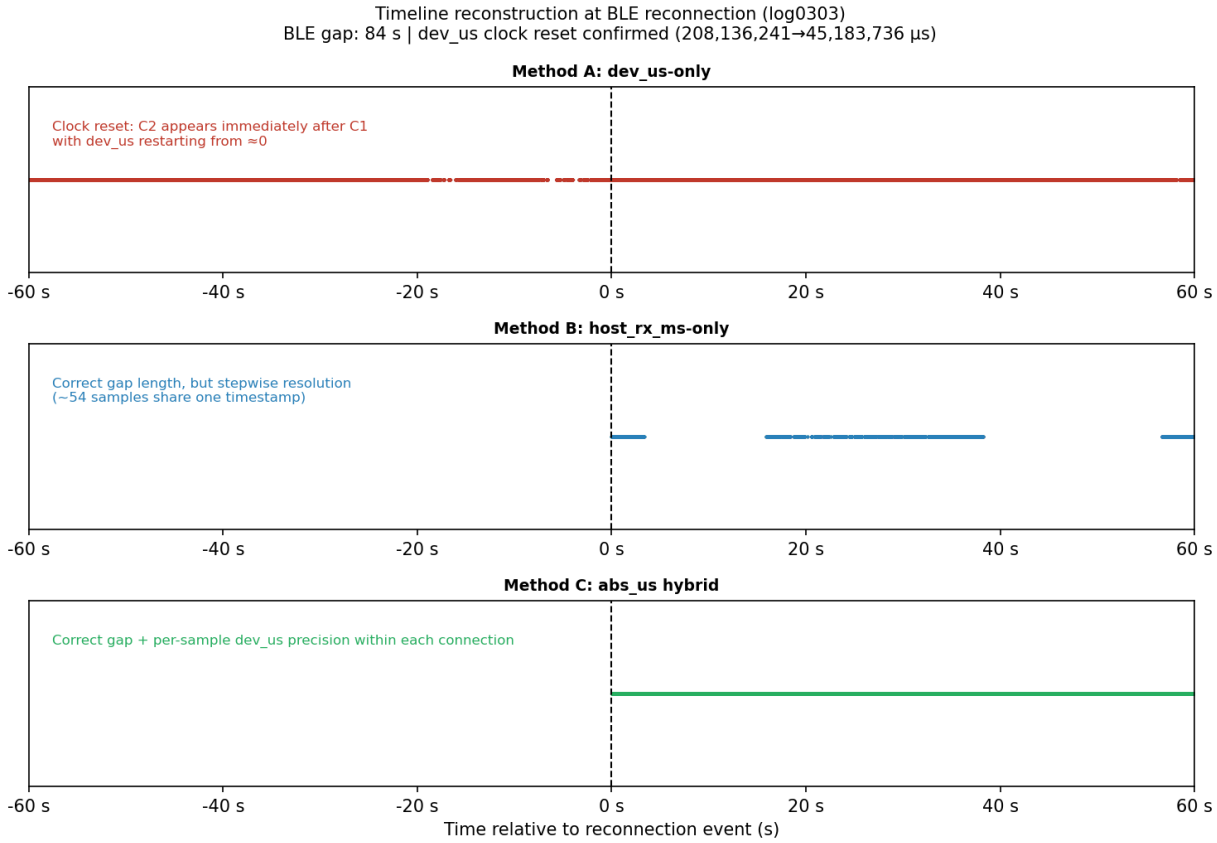
Method B (*host_rx_ms*-only): Host wall-clock milliseconds, continuous across reconnects, but assigns the same timestamp to all $\sim$54 samples in one BLE notification batch. Per-sample timing loss: p50 = 13.5 ms, p95 = 52.5 ms, p99 = 81.0 ms.

Method C (*abs_us* hybrid, this work): *dev_us* per-sample precision within connection (± 25 μs), *host_rx_ms* gap bridging across reconnects. Cross-connection accuracy bounded by BLE notification interval (p99 $\approx$ 170 ms).

BLE notification interval characterization across 7.10 million samples (log0305, 59.2 min): p50 = 9 ms, p95 = 105 ms, p99 = 170 ms; mean 53.8 samples per notification.



(**a**) BLE notification interval CDF (log0305, 59.2 min). Left panel shows CDF of inter-notification intervals; right panel shows a 5-second window with *host_rx_ms* batch granularity (staircase) vs. *dev_us* per-sample precision (smooth line).



(**b**) Three-method timeline comparison at BLE reconnection (log0303, 84.0-s gap). Method A: clock reset causes C2 to appear immediately after C1. Method B: correct gap with batch-level granularity. Method C (*abs_us* hybrid): correct gap with per-sample precision.

Figure 4. Quantitative *abs_us* timeline validation. (**a**) BLE notification interval characterization. (**b**) Method comparison at reconnect.

5.5. Pogo-Pin Ghost-Signal Field Failure

Before the GPIO 32 solder bypass, the RAW channel exhibited a ghost signal with Pearson $r = 0.998$ vs. ENV, consistent with ADC multiplexer charge sharing or floating-input coupling. After bandpass filtering (50–450 Hz), the ghost signal flatlined, causing silent loss of usable RAW spectral EMG. Post-bypass validation confirmed independent

AC EMG oscillation (std $>$ 100 ADC counts during contraction) and ENV–RAW correlation dropping to $r = -0.013$ for filtered signals.

Generalization. Any wearable routing a high-impedance analog signal through a non-soldered contact (pogo pin, snap, spring contact) and sampling on a multiplexed ADC is vulnerable to this failure mode. Diagnostic heuristic: Pearson $r > 0.95$ between two channels that should be electrically independent, with near-zero variance on one, indicates crosstalk or a floating pin. Post-filter flatline silently removes usable spectral data.

6. Results

6.1. Three-Phase MDF Trajectory

Figure 5 shows the central empirical finding: a three-phase MDF trajectory rather than a simple monotonic decline.

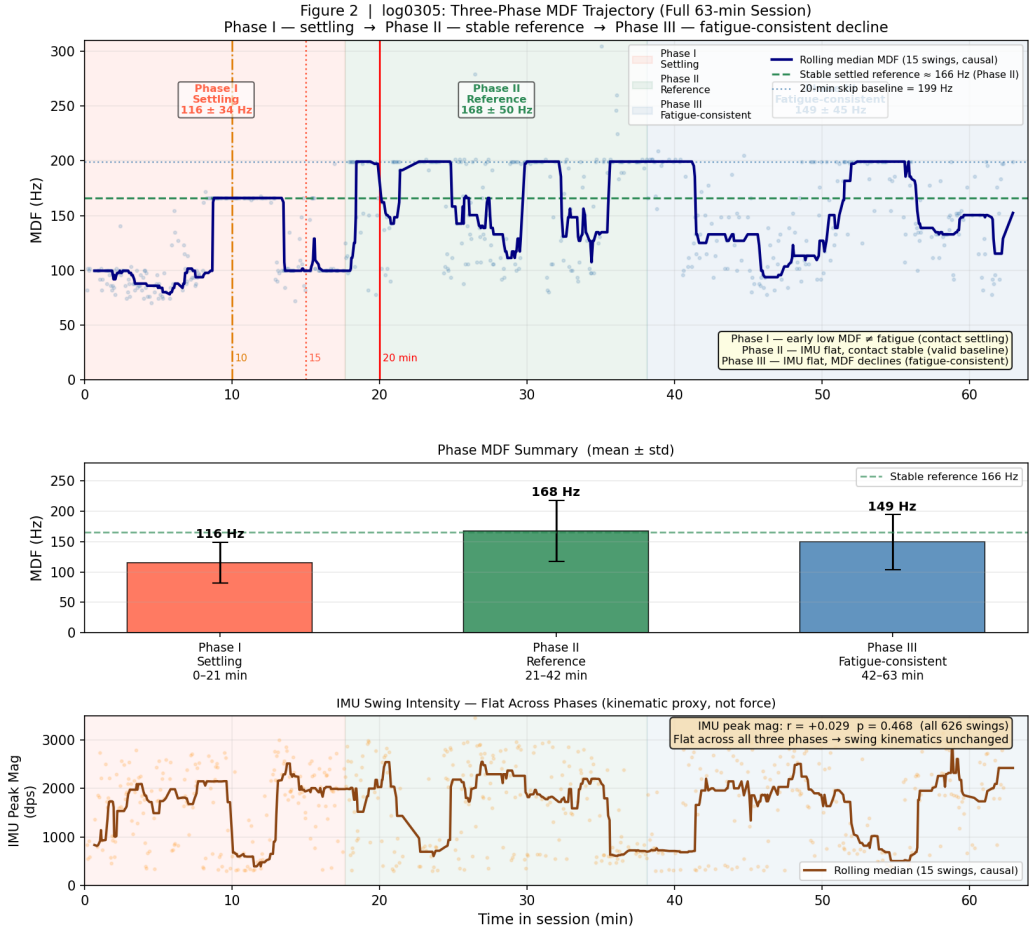


Figure 5. Three-phase MDF trajectory in log0305 (63.1 min, 626 swings). Phase I (settling, 0–21 min): MDF 115.7 ± 33.5 Hz. Phase II (stable reference, 21–42 min): MDF 167.6 ± 50.1 Hz. Phase III (fatigue-consistent, 42–63 min): MDF 149.5 ± 45.1 Hz. IMU swing intensity remains flat across all phases ($r = +0.029$, $p = 0.468$). Warmup markers at 10/15/20 min shown.

Phase I (settling): MDF 115.7 ± 33.5 Hz—substantially below the settled reference. Phase II (reference): MDF 167.6 ± 50.1 Hz—post-settling stable baseline. Phase III (fatigue-consistent): MDF 149.5 ± 45.1 Hz—decline after the settled reference. Early and late low-MDF regions have different causes: early suppression reflects contact settling (MDF subsequently recovered while IMU remained flat); late decline is fatigue-consistent.

6.2. Warmup-Skip Sensitivity

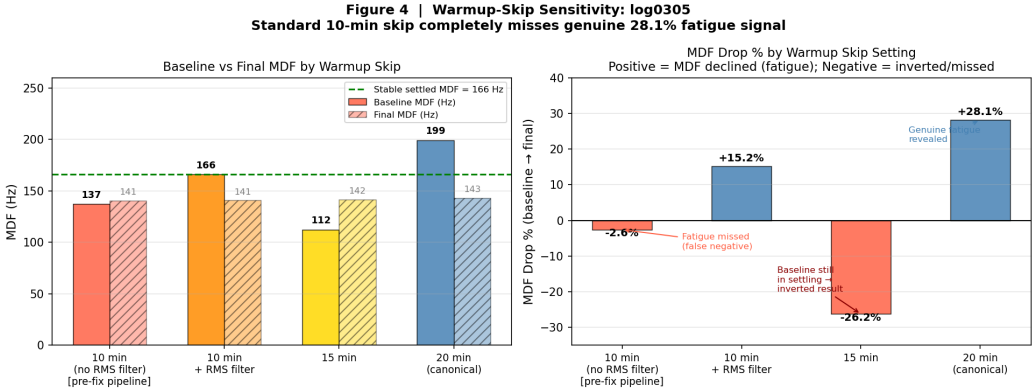


Figure 6. Warmup-skip sensitivity analysis. Left: baseline MDF per skip condition vs. stable settled reference (167.6 Hz). Right: MDF change (%) per condition. A 10-min skip produces −2.6% (fatigue missed); 15-min produces −26.2% (inverted); 20-min produces +28.1% MDF decline (fatigue-consistent).

A 10-min skip produces −2.6% MDF change (fatigue signal missed entirely). A 15-min skip produces −26.2% (inverted—baseline still in settling phase). A 20-min skip reveals +28.1% MDF decline and +17.3% MNF decline. The non-monotone behavior (15-min worse than 10-min) reflects the RMS-based baseline selector catching swings from the steepest part of the settling transition.

6.3. IMU-Conditioned Validity

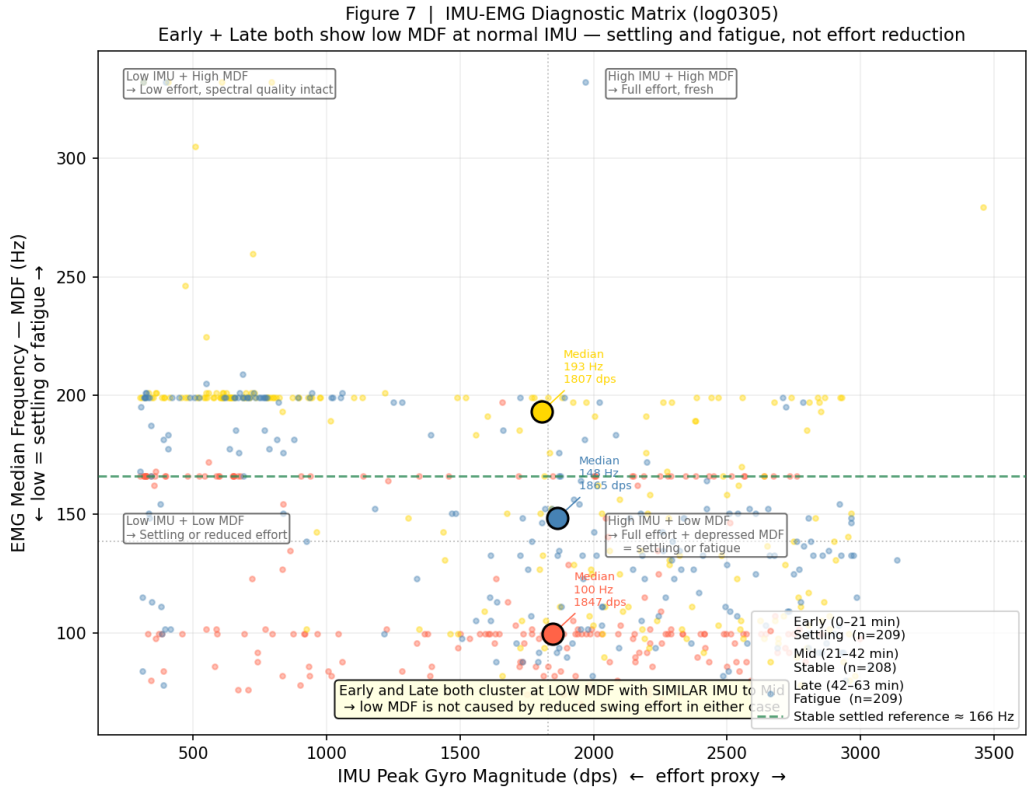


Figure 7. IMU–EMG diagnostic matrix (log0305, 626 swings). IMU peak gyro magnitude (kinematic proxy) vs. MDF, colored by session third. Early and Late swings cluster at lower MDF despite similar IMU intensity. $r = +0.029$, $p = 0.468$.

All three session thirds show similar IMU intensity ($r = +0.029$, $p = 0.468$), confirming that reduced gross swing kinematics is an unlikely explanation for either Phase I suppression or Phase III decline.

6.4. Contact-Quality Evolution

In log0305, contact quality increased from early to mid-session (0.36 ± 0.28 to 0.49 ± 0.47) and declined late (0.25 ± 0.32), consistent with electrode settling followed by activation-amplitude changes (Figure 8).

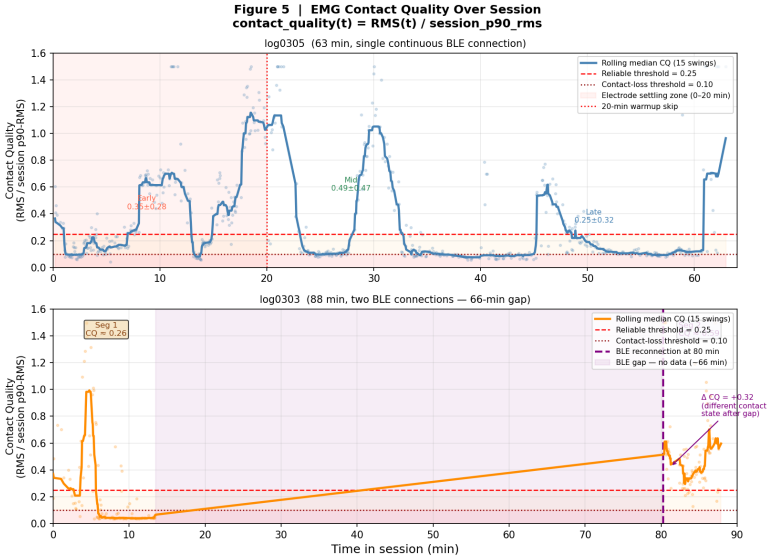


Figure 8. Contact quality evolution (log0305). RMS-normalized metric increases during Phase I–II settling and shows late-session decline.

In log0303, contact quality differed substantially between segments (Segment 1: 0.26 vs. Segment 2: 0.59), supporting that the full-session MDF drop reflected a contact state change across the gap rather than a continuous physiological trend.

6.5. Session-Gap Artifact

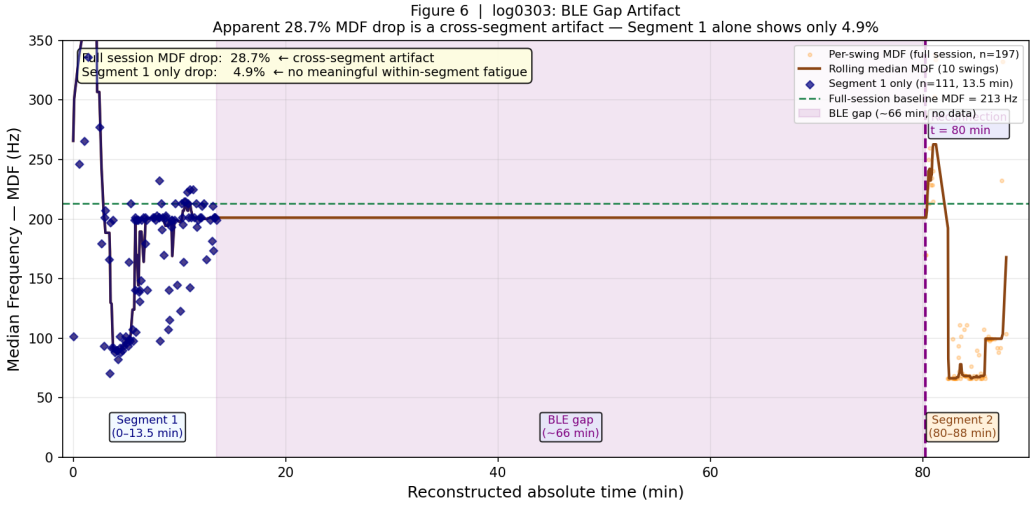


Figure 9. log0303 session-gap artifact. Full session (orange): apparent 28.7% MDF drop. First continuous BLE segment alone (navy): 4.9% change. An 84-second BLE reconnection gap (66 min apparent practice gap) separates the two segments (purple shading).

The full-session 28.7% MDF drop compares Segment 1 against Segment 2 across an 84-second BLE gap representing a 66-min interruption during which electrode contact state and player condition may have changed. Segment 1 alone shows 4.9%—below any meaningful fatigue threshold for 13.5 min of play.

6.6. Ball-Impact Masking Impact on MDF Estimates

Figure 10 shows the per-swing MDF comparison between full-window and pre-impact Approach B windowing.

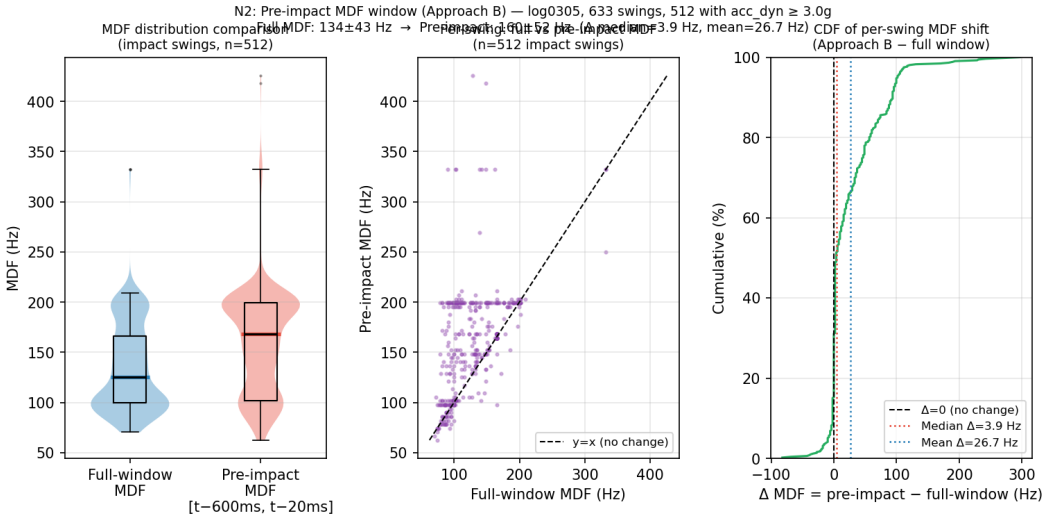


Figure 10. Pre-impact MDF window analysis (log0305, 633 swings, 512 with $a_{\text{dyn}} \geq 3.0$ g). **(a)** Violin + box comparison of full-window vs. pre-impact MDF distributions. **(b)** Per-swing scatter (full vs. pre-impact MDF). **(c)** CDF of ΔMDF = pre-impact − full-window.

Of 633 total swings, 512 (81%) showed ball impact ($a_{\text{dyn}} \geq 3.0$ g; mean peak 11.1 g). Full-window MDF for impact swings: 133.7 ± 43.4 Hz (median 125.0 Hz). Pre-impact MDF (Approach B): 160.4 ± 52.0 Hz (median 168.0 Hz). Δ = pre-impact − full: mean = +26.7 Hz, std = 44.5 Hz, median = +3.9 Hz; p75 = +46.9 Hz, p95 = +101.6 Hz. The distribution is right-skewed: 25% of swings show no shift, 50% show positive.

The pre-impact window captures the active muscle contraction phase before mechanical disturbance, resulting in systematically higher MDF. The magnitude (+26.7 Hz mean) confirms that window selection is physiologically meaningful for impact-bearing swings and that IMU events enable principled temporal partitioning of EMG analysis windows.

6.7. Multi-Subject Results

[TODO: Data collection pending. See N3 in journal_plan.md.]

7. Discussion

7.1. Platform–Methodology Coupling

The platform and methodology are tightly coupled: neither is useful without the other. The spectral fatigue analysis (Sections 6) requires: (i) RAW EMG access (not available through the pogo-pin path; required the solder bypass); (ii) *abs_us* timeline (required for correct session-gap artifact detection; Method A would silently merge the two log0303 segments); (iii) IMU-as-context (required for swing detection, validity conditioning, and pre-impact window masking); and (iv) 2000 Hz EMG (required for 50–450 Hz bandpass and Welch PSD). These dependencies motivate the combined work.

7.2. Three-Phase Trajectory as a General Framework

The three-phase settling–reference–fatigue trajectory is predicted by dry-electrode physics (Section 4) and should appear in any no-gel field sEMG deployment with sufficient session length. Phase I duration depends on sweat rate and ambient conditions; the warmup-skip sensitivity protocol (Section 4) provides a session-specific robustness check regardless of phase boundaries.

7.3. Pogo-Pin Failure Mode Generalization

ADC multiplexer charge sharing generalizes to capacitor microphones, piezoelectric sensors, and other high-impedance biosignal sources routed through multiplexed ADCs via non-soldered contacts. The diagnostic heuristic ($r > 0.95$ between independent channels with near-zero variance on one) is applicable to any such wearable.

7.4. Practical Recommendations

Based on these results, we recommend: (1) use $\geq$20 min warmup exclusion for no-gel forearm snap-electrode deployments; (2) report warmup-skip sensitivity as a standard robustness audit; (3) use IMU swing intensity as an independent validity channel; (4) audit BLE session continuity before computing cross-session baselines; (5) verify RAW EMG access via pogo-pin inspection before field deployment; (6) use pre-impact MDF windows when IMU-detected ball contact events are present.

7.5. Limitations

Currently reported results cover a single subject (pilot study); multi-subject generalization is pending (Section 6.7). No MVC, RPE, grip-force, or ball-speed ground truth is available—findings are fatigue-consistent rather than physiologically confirmed. The 20-min warmup skip is empirically selected for this session; the contact-quality metric is not validated against direct impedance. The combined BLE data rate ($\sim$12.5 KB/s) is near BLE 4.2 capacity at MTU 244; adding sensors requires rate reduction or BLE 5.0 migration.

8. Conclusions

The platform–methodology coupling demonstrated here enables field sEMG fatigue monitoring that would not be possible with either component alone. The $99 ESP32 wearable provides $\pm$25 μs timestamp accuracy (99.03% over 2.3 million samples), a self-describing TLV protocol surviving three FSR revisions, zero detected sample drops during 150+ min of field sessions, and documented pogo-pin failure diagnosis and repair.

The methodology reveals that no-gel field sEMG fatigue monitoring is fundamentally a trustworthiness problem. The same session changed from $-$2.6% to $+$28.1% MDF drop depending solely on the warmup exclusion window, demonstrating that baseline selection is a first-order determinant of fatigue interpretation. The IMU, as an independent kinematic context channel, confirmed stable swing intensity ($r = +0.029$, $p = 0.468$) across all phases, attributing Phase I suppression to electrode settling and Phase III decline to fatigue-consistent spectral shift. A BLE reconnection gap created an apparent 28.7% spurious fatigue signal that collapsed to 4.9% for the first continuous segment—demonstrating the necessity of continuity auditing. Pre-impact MDF windowing (Approach B) shows a mean $+$26.7 Hz shift vs. full-window MDF, confirming that temporal partitioning guided by IMU events is physiologically meaningful.

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Institutional Review Board Statement: Data were collected using non-invasive wearable sEMG/IMU sensors during normal tennis practice. Written informed consent was obtained from a parent or legal guardian; participant assent was obtained where applicable. Data were anonymized before analysis.

Informed Consent Statement: Written informed consent was obtained from all subjects involved in the study.

Data Availability Statement: Analysis scripts are available at https://github.com/[TODO:addrepo]. Raw session log files are available upon reasonable request.

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